a. We have $\phi((a+bi)+(c+di)) = \phi((a+c)+(b+d)i) = \begin{bmatrix} a+c & -(b+d) \\ b+d & a+c \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \phi(a+bi)+\phi(c+di)$.

b. We have $\phi((a+bi)\cdot(c+di)) = \phi((ac-bd)+(ad+bc)i) = \begin{bmatrix} ac-bd & -(ad+bc) \\ ad+bc & ac-bd \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \phi(a+bi)\cdot\phi(c+di)$.

34. Let the set be $\{a, b\}$. We need to decide whether interchanging the names of the letters everywhere in the table and then writing the table again in the order a first and b second gives the same table or a different table. The same table is obtained if and only if in the body of the table, diagonally opposite entries are different. Four such tables exist, since there are four possible choices for the first row; Namely, the tables

$*$	a	b
a	a	a
b	b	b

$*$	a	b
a	a	b
b	a	b

$*$	a	b
a	b	a
b	b	a

and

$*$	a	b
a	b	b
b	a	a

.

The other 12 tables can be paired off into tables giving the same algebraic structure. One table of each pair is listed below. The number of different algebraic structures is therefore $4 + 12/2 = 10$.

$*$	a	b
a	a	a
b	a	a

$*$	a	b
a	a	a
b	a	b

$*$	a	b
a	a	a
b	b	a

$*$	a	b
a	a	b
b	a	a

$*$	a	b
a	b	a
b	a	a

$*$	a	b
a	a	b
b	b	a

4. Groups

1. No. G_3 fails. **2.** Yes **3.** No. G_1 fails. **4.** No. G_3 fails. **5.** No. G_1 fails.

6. No. G_2 fails.

7. The group $\langle U_{1000}, \cdot\rangle$ of solutions of $z^{1000} = 1$ in $\mathbb{C}$ under multiplication has 1000 elements and is abelian.

8.

$\cdot_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

9. Denoting the operation in each of the three groups by $*$ and the identity element by e for the moment, the equation $x*x*x*x = e$ has four solutions in $\langle U, \cdot\rangle$, one solution in $\langle \mathbb{R}, +\rangle$, and two solutions in $\langle \mathbb{R}^*, \cdot\rangle$.

10. **a.** *Closure:* Let nr and ns be two elements of $n\mathbb{Z}$. Now $nr + ns = n(r+s) \in n\mathbb{Z}$ so $n\mathbb{Z}$ is closed under addition.

Associative: We know that addition of integers is associative.

Identity: $0 = n0 \in n\mathbb{Z}$, and 0 is the additive identity element.

Inverses: For each $nm \in n\mathbb{Z}$, we also have $n(-m) \in n\mathbb{Z}$ and $nm + n(-m) = n(m-m) = n0 = 0$.

b. Let $\phi : \mathbb{Z} \to n\mathbb{Z}$ be defined by $\phi(m) = nm$ for $m \in \mathbb{Z}$. Clearly ϕ is one to one and maps $\mathbb{Z}$ onto $n\mathbb{Z}$. For $r, s \in \mathbb{Z}$, we have $\phi(r+s) = n(r+s) = nr + ns = \phi(r) + \phi(s)$. Thus ϕ is an isomorphism of $\langle \mathbb{Z}, + \rangle$ with $\langle n\mathbb{Z}, + \rangle$.

11. Yes, it is a group. Addition of diagonal matrices amounts to adding in $\mathbb{R}$ entries in corresponding positions on the diagonals, and that addition is associative. The matrix with all entries 0 is the additive identity, and changing the sign of the entries in a matrix yields the additive inverse of the matrix.

12. No, it is not a group. Multiplication of diagonal matrices amounts to muliplying in $\mathbb{R}$ entries in corresponding positions on the diagonals. The matrix with 1 at all places on the diagonal is the identity element, but a matrix having a diagonal entry 0 has no inverse.

13. Yes, it is a group. See the answer to Exercise 12.

14. Yes, it is a group. See the answer to Exercise 12.

15. No. The matrix with all entries 0 is upper triangular, but has no inverse.

16. Yes, it is a group. The sum of upper-triangular matrices is again upper triangular, and addition amounts to just adding entries in $\mathbb{R}$ in corresponding positions.

17. Yes, it is a group.

Closure: Let A and B be upper triangular with determinant 1. Then entry c_{ij} in row i and column j in $C = AB$ is 0 if $i > j$, because for each product $a_{ik}b_{kj}$ where $i > j$ appearing in the computation of c_{ij}, either $k < i$ so that $a_{ik} = 0$ or $k \geq i > j$ so that $b_{kj} = 0$. Thus the product of two upper-triangular matrices is again upper triangular. The equation $\det(AB) = \det(A) \cdot \det(B)$, shows that the product of two matrices of determinant 1 again has determinant 1.

Associative: We know that matrix multiplication is associative.

Identity: The $n \times n$ identity matrix I_n has determinant 1 and is upper triangular.

Inverse: The product property $1 = \det(I_n) = \det(A^{-1}A) = \det(A^{-1}) \cdot \det(A)$ shows that if $\det(A) = 1$, then $\det(A^{-1}) = 1$ also.

18. Yes, it is a group. The relation $\det(AB) = \det(A) \cdot \det(B)$ show that the set of $n \times n$ matrices with determinant ± 1 is closed under multiplication. We know matrix multiplication is associative, and $\det(I_n) = 1$. As in the preceding solution, we see that $\det(A) = \pm 1$ implies that $\det(A^{-1}) = \pm 1$, so we have a group.

19. a. We must show that S is closed under $*$, that is, that $a+b+ab \neq -1$ for $a, b \in S$. Now $a+b+ab = -1$ if and only if $0 = ab + a + b + 1 = (a+1)(b+1)$. This is the case if and only if either $a = -1$ or $b = -1$, which is not the case for $a, b \in S$.

b. *Associative:* We have

$$a * (b * c) = a * (b + c + bc) = a + (b + c + bc) + a(b + c + bc) = a + b + c + ab + ac + bc + abc$$

and

$$(a * b) * c = (a + b + ab) * c = (a + b + ab) + c + (a + b + ab)c = a + b + c + ab + ac + bc + abc.$$

Identity: 0 acts as identity elemenr for $*$, for $0 * a = a * 0 = a$.

Inverses: $\frac{-a}{a+1}$ acts as inverse of a, for

$$a * \frac{-a}{a+1} = a + \frac{-a}{a+1} + a\frac{-a}{a+1} = \frac{a(a+1) - a - a^2}{a+1} = \frac{0}{a+1} = 0.$$

c. Because the operation is commutative, $2 * x * 3 = 2 * 3 * x = 11 * x$. Now the inverse of 11 is -11/12 by Part(**b**). From $11 * x = 7$, we obtain

$$x = \frac{-11}{12} * 7 = \frac{-11}{12} + 7 + \frac{-11}{12}7 = \frac{-11 + 84 - 77}{12} = \frac{-4}{12} = -\frac{1}{3}.$$

20.

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table I

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

Table II

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Table III

Table I is structurally different from the others because every element is its own inverse. Table II can be made to look just like Table III by interchanging the names a and b everywhere to obtain

	e	b	a	c
e	e	b	a	c
b	b	e	c	a
a	a	c	b	e
c	c	a	e	b

and rewriting this table in the order e, a, b, c.

a. The symmetry of each table in its main diagonal shows that all groups of order 4 are commutative.

b. Table III gives the group U_4, upon replacing e by 1, a by i, b by -1, and c by $-i$.

c. Take $n = 2$. There are four 2×2 diagonal matrices with entries ± 1, namely

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \text{ and } C = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

If we write the table for this group using the letters E, A, B, C in that order, we obtain Table I with the letters capitalized.

21. A binary operation on a set $\{x, y\}$ of two elements that produces a group is completely determined by the choice of x or y to serve as identity element, so just 2 of the 16 possible tables give groups. For a set $\{x, y, z\}$ of three elements, a group binary operation is again determined by the choice x, y, or z to serve as identity element, so there are just 3 of the 19,683 binary operations that give groups. (Recall that there is only one way to fill out a group table for $\{e, a\}$ and for $\{e, a, b\}$ if you require e to be the identity element.)

22. The orders $G_1G_3G_2, G_3G_1G_2$, and $G_3G_2G_1$ are not acceptable. The identity element e occurs in the statement of G_3, which must not come before e is defined in G_2.

23. Ignoring spelling, punctuation and grammar, here are some of the mathematical errors.

a. The statement "x = identity" is wrong.

b. The identity element should be e, not (e). It would also be nice to give the properties satisfied by the identity element and by inverse elements.

c. Associativity is missing. Logically, the identity element should be mentioned before inverses. The statement "an inverse exists" is not quantified correctly: for each element of the set, an inverse exists. Again, it would be nice to give the properties satisfied by the identity element and by inverse elements.

d. Replace " such that for all $a, b \in G$" by " if for all $a \in G$". Delete "under addition" in line 2. The element should be e, not $\{e\}$. Replace "$= e$" by "$= a$" in line 3.

24. We need only make a table that has e as an identity element and has an e in each row and each column of the body of the table to satisfy axioms G_2 and G_3. Then we make some row or column contain some element twice, and it can't be a group, so G_1 must fail.

$*$	e	a	b
e	e	a	b
a	a	e	b
b	b	a	e

25. F T T F F T T T F T

26. Multiply both sides of the equation $a * b = a * c$ on the left by the inverse of a, and simplify, using the axioms for a group.

27. Show that $x = a' * b$ is a solution of $a * x = b$ by substitution and the axioms for a group. Then show that it is the only solution by multiplying both sides of the equation $a * x = b$ on the left by a' and simplifying, using the axioms for a group.

28. Let $\phi : G \to G'$ be a group isomorphism of $\langle G, * \rangle$ onto $\langle G', *' \rangle$, and let $a, a' \in G$ such that $a * a' = e$. Then $\phi(e) = \phi(a * a') = \phi(a) *' \phi(a')$. Now $\phi(e)$ is the identity element of G' by Theorem 3.14. Thus the equation $\phi(a) *' \phi(a') = \phi(e)$ shows that $\phi(a)$ and $\phi(a')$ are inverse pairs in G', which was to be shown.

29. Let $S = \{x \in G \mid x' \neq x\}$. Then S has an even number of elements, because its elements can be grouped in pairs x, x'. Because G has an even number of elements, the number of elements in G but not in S (the set $G - S$) must be even. The set $G - S$ is nonempty because it contains e. Thus there is at least one element of $G - S$ other than e, that is, at least one element other than e that is its own inverse.

30. **a.** We have $(a * b) * c = (|a|\, b) * c\ |(|a|b)|c = |\, ab|c$. We also have a $*(b * c) = a * (|\, b|c) = |a||\, b|c = |\, ab|c$, so $*$ is associative.

b. We have $1 * a = |1|\, a = a$ for all $a \in \mathbb{R}^*$ so 1 is a left identity element. For $a \in \mathbb{R}^*, 1/|\, a|$ is a right inverse.

c. It is not a group because both 1/2 and -1/2 are right inverse of 2.

d. The one-sided definition of a group, mentioned just before the exercises, must be all left sided or all right sided. We must not mix them.

31. Let $\langle G, * \rangle$ be a group and let $x \in G$ such that $x * x = x$. Then $x * x = x * e$, and by left cancellation, $x = e$, so e is the only idempotent element in a group.